Default Final Project

NOVA50101 Introduction to Artificial Intelligence
for Industrial AI

Deadline: Sunday, May 10, 2026, 11:59 PM

1 Introduction

In the final project, you will explore an interesting machine learning problem using the machine learning and deep learning algorithms learned in class along with **Kaggle** datasets. Each team should consist of 3–4 members, and if the team has fewer than 3 members, approval from the professor is required. You can choose the dataset freely from **Kaggle** datasets, but you should carefully check whether it is appropriate for your project and this course. The complexity of the chosen dataset will determine the difficulty of the project. If you are unsure how to start, please ask the TA for guidance.
Once you form a team, send the team leader's name and members' names to the TA, with professor approval (for 2-member) if it exists. If the team has fewer than 3 members, or if no email is sent, teams will be assigned randomly. (e.g., for not approved 2-member team, an attempt will be made to add one or two random members, but in some cases random reassignment may occur.) The deadline to email the TA for team assignment is **May 3**. Team numbers will be assigned by the TA and announced on Blackboard on **May 4**. Please refer to it.

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Project Proposal and Topic Selection. Only the team leader should submit the proposal. The final team compositions will be announced later on Blackboard. The first task of the final project is to select a topic. Most students will undertake one of three types of projects:

- **Application Project.** This is the most common type. You will select an application area of interest and explore how learning algorithms can be best applied to solve a problem within it.
- **Algorithmic Project.** You will pick a problem or a family of problems and develop a new learning algorithm or a new variant of an existing algorithm to solve it.

- **Theoretical Project.** You will prove interesting and non-trivial properties of a new or existing learning algorithm. (This is often quite difficult, so purely theoretical projects are rare.)

Additionally, some projects may combine elements of application, algorithmic, and theoretical work. Many of the best class projects come from students choosing an application area they are passionate about or a subfield of machine learning they want to explore further. So, please choose something that you can be excited about, and feel free to propose ambitious ideas that interest you. If you have any questions about your choice, you are always welcome to consult with the TAs. If the dataset you choose from Kaggle requires significant preprocessing, or if you need to collect it yourself, please keep in mind that this is only part of the expected project work but can often take a significant amount of time. We still expect a solid discussion of methodology and results, so please pace your project appropriately.

Notes on Specific Project Types. Since this course covers many concepts beyond deep learning, if you decide to do a deep learning project, you must also incorporate other topics from the class. For example, you could set up logistic regression and SVM baselines or perform data analysis using unsupervised learning methods covered in the course. Training deep learning models can be very time-consuming, so ensure you have access to the necessary computing resources. You can use environments like Google Colab for simple GPU access.

Pre-processed Datasets. While we don't want you to spend too much time collecting raw data, the process of examining and visualizing data, trying different types of pre-processing, and performing error analysis is a crucial part of machine learning. Therefore, we encourage you to perform data exploration and analysis to familiarize yourself with the problem, even with a prepared dataset.

Reproducing Results. Reproducing a paper's results can be a great learning experience, especially if there are results to compare against. However, instead of simply reproducing the work, you should try to apply the technique to a different application or perform an analysis of how each component of the model contributes to the final performance. In other words, while your project does not need to be entirely new, it should not be a simple replication of prior work.

2 Evaluation

The final project will be evaluated based on the following 3 main criteria:

1. **Technical soundness (70%).** This includes the proposed method, initial experimental results, technical validity, originality, insight from results, and appropriateness of project scale. In addition, clarity and effectiveness in presenting the project context, methods, and results will be assessed.

2. **Limitation & Future Work (25%).** This covers the design of future experiments and extensions of the algorithm. Based on the results of your project, each team should discuss its limitations and future directions.

3. **Activities (5%).** Teams should submit an appendix showing both the original and revised activity plans, along with the contributions of each member.

The originality of your project will be evaluated by comparing it with the most upvoted codes on your dataset. Achieving better performance is desirable but not required, and you may also further develop the related codes with your own ideas.

3 Proposal

The final project proposal should be a maximum of two pages and include the following:

1. Team number, members, and project title
2. Dataset information you will use for project and baselines using your chosen dataset
3. Brief descriptions of what the project is and how you plan to approach it
4. Roles of team members

The dataset information should include the name, the **Kaggle** URL, and the dataset size and type. For baselines, submit the URLs of the top 3 public codes that achieved the best performance on the chosen dataset. If no public code is available for the dataset, you do not need to submit baselines' URLs.
The final project proposal is due on **May 10** and should be submitted on Blackboard. Feedback will be given. If you plan to use a deep learning algorithm not covered in class, please inform the TA or the professor in advance.